

BRAC University

Department of Computer Science and Engineering

CSE111: Programming Language II

Examination: Quiz #2

Date: ____ / ____ / 2022

Semester: Summer 2022

Time: 40 Minutes

ID: _____	Name: (Please write in CAPITAL LETTERS)	Section: 42
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- Read questions and allocate time for each question carefully. Answer all the questions.
- Understanding the question is part of the exam, please do not ask questions.

1. Design a ExamTracker class and implement the following methods:

- Constructor - `__init__(...)` - Method (2 Marks)
- `add_topics(...)` Method (5 Marks)
- `update_date(...)` Method - [OPTIONAL] (4 Marks)
- `print_details()` Method (2 Marks)
- `get_details()` Method (5 Marks)

To generate the given output from the driver code. [CO1, CO2, CO4, CO6]

Driver Code		Output
1	<code>cse111_quiz = ExamTracker("CSE111", "Quiz")</code>	<pre>New ExamTracker has been created... CSE111 exam date for Quiz - 1 has been changed from 15/06/2022 to 20/06/2022 ===== Course: CSE111 Exam Title: Quiz Modules: Quiz - 1 Exam Date: 20/06/2022 - String Manipulations - List and Tuple - Dictionary - Functions</pre>
2	<code>cse111_quiz.add_topics("String Manipulations", "List and Tuple", module="1")</code>	
3	<code>cse111_quiz.add_topics("Dictionary", "Functions", exam_date="15/06/2022", module="1")</code>	
4	<code>cse111_quiz.update_date("20/06/2022", module="1")</code>	
5	<code>cse111_quiz.print_details()</code>	
6	<code>cse111_mid = ExamTracker("CSE111", "Midterm", "19/07/2022")</code>	<pre>New ExamTracker has been created... ===== Course: CSE111 Exam Title: Midterm Modules: No modules have been created yet. ===== Course: CSE111 Exam Title: Midterm Modules: Midterm - default Exam Date: 19/07/2022 - Up to Quiz 2</pre>
7	<code>print(cse111_mid.get_details())</code>	
8	<code>cse111_mid.add_topics("Up to Quiz 2")</code>	
9	<code>cse111_mid.print_details()</code>	

2. Trace the following code and show the **complete** tracing table. [CO2]

(6 Marks)

1	class Msg:
2	def __init__(self, content):
3	self.content = content
4	class Quiz2:
5	def __init__(self):
6	self.a = Msg(_X_) # X: Last_two digits of your ID
7	self.b = Msg(_Y_) # Y: First_digit * Last_digit of your ID
8	self.c = Msg(_Z_) # Z: Last_digit of your ID + Current_month_number
9	self.x, self.y, self.sum = 2, 2, 0
10	def method_a(self):
11	x, y = self.c.content, self.a.content
12	self.a.content = self.x + self.method_b(self.b) + self.x + y
13	x = self.a.content + x
14	y = self.y + self.method_b() + self.b.content
15	self.sum = self.a.content + self.y
16	print(x, y, self.sum, sep=" ")
17	def method_b(self, mg1=None):
18	if mg1 != None:
19	sum = self.x * self.y ** 2 + self.sum
20	self.a = mg1
21	return sum
22	else:
23	self.b = self.c
24	self.y += 1
25	return self.sum + self.y ** 2
26	
27	q = Quiz2()
28	q.method_a()
29	q.method_a()
30	q.method_a()